

TUNING RATIONALE - TEAM 05

MC3113 Lab 2 | 10 March 2026

Team Members:

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Controller Type: PID

ITERATION LOG:

Iter	Kp	Ki	Kd	base	IAE(m.s)	Distance(m)	Energy	R-TRACK-02 met?
0(P-only)	-1.0	0	0	0.4	18.2391	4.5	72.5	Not met
1 (PID Baseline)	-2.0	-0.30	-0.05	0.4	13.2296	7.5	86.5	Not met
2	-3.0	-0.20	-0.08	0.6	13.3223	10.3	110.0	Not met
3	-2.5	-0.15	-0.10	0.6	5.9722	31.0	107.0	Not met
4	-3.0	-0.20	-0.12	0.65	3.9299	42.2	116.8	Not met
5	-3.5	-0.25	-0.15	0.7	3.4103	46.2	125.1	Not met
6	-4.0	-0.30	-0.18	0.72	3.2635	47.2	127.8	Not met
7	-3.2	-0.20	-0.25	0.68	2.4216	46.9	122.0	Not met
8	-2.5	-0.10	-0.30	0.60	2.3057	41.7	107.9	Not met
9	-1.8	-0.05	-0.40	0.55	2.0490	38.6	98.9	Not met
10	-1.5	-0.02	-0.45	0.52	1.9450	36.6	93.7	Not met
11	-1.3	-0.01	-0.50	0.50	1.8941	35.1	90.5	Not met
12	-1.2	0	-0.52	0.48	1.9488	33.7	87.3	Not met
13	-0.8	0	-0.60	0.40	2.0624	28.0	76.3	Not met
14(Final)	-0.6	0	-0.80	0.32	2.1222	21.9	74.0	Not met

FINAL CONFIGURATION (Iteration 14):

$K_p = -0.6$, $K_i = 0$, $K_d = -0.80$, base = 0.32

PERFORMANCE SUMMARY:

Metric	Target	Achieved	Status
IAE	≤ 0.03	2.1222	Not met (CA3 target)
Track progress	≥ 48 m	21.9 m	Not met
Energy proxy	≤ 60	74.0	Not met

KEY DESIGN DECISIONS:

Systematically reduced energy by lowering K_p , removing K_i , and increasing K_d .

Best energy achieved at Iteration 14 (74.0) while maintaining IAE under 2.2.

Trade-off: Lower energy resulted in reduced track progress.

CONCLUSION:

The controller meets the lab requirements for systematic tuning with 14 iterations.

R-TRACK-02 (IAE ≤ 0.03) is not met in this lab but will be targeted for CA3.